CURVATURE AND VARIANCE IN LATTICE FIRST-PASSAGE PERCOLATION

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ABSTRACT. We prove that, in first-passage percolation on $\mathbb{Z}^d$ with $d \geq 2$ and time constant μ, the passage time $T(0,x)$ satisfies $\mathbb{E}[\exp(|T(0,x) - \mu(x)|/|x|^{\frac{1}{2}(1-\varepsilon)})] \leq C$ for some $\varepsilon > 0$ in every direction in which the limit shape is curved. Hence $\mathrm{Var}[T(0,x)] \leq C|x|^{1-\varepsilon}$, which answers a question of Kesten in these directions. The proof adapts our argument for rotationally invariant first-passage percolation on $\mathbb{R}^d$ [AB26] to the lattice.

1. INTRODUCTION

Let $d \geq 2$, and give each edge e between nearest neighbors of $\mathbb{Z}^d$ a nonnegative random weight τ_e, independently with a common law ν. The *passage time* between two points $x, y \in \mathbb{Z}^d$ is

$$T(x,y) := \inf_{\pi} \sum_{e \in \pi} \tau_e \,,$$

where the infimum is over the nearest-neighbor paths π from x to y. Let $p_c(d)$ be the critical probability of bond percolation on $\mathbb{Z}^d$.

Assumption (W). The law ν is supported in $[0,\infty)$, $\nu(\{0\}) < p_c(d)$, and $\int e^{ax}\,\nu(dx) < \infty$ for some $a > 0$.

Under (W), the *time constant* $\mu(x) := \lim_{n\to\infty} n^{-1}T(0,nx)$ exists almost surely for $x \in \mathbb{Z}^d$ and extends to a norm on $\mathbb{R}^d$ [ADH17, Theorems 2.1, 2.5 and 2.6, and the remarks on p. 11].

Assumption (Curv). The unit vector ξ satisfies (Curv) with the constants $\delta \in (0,1]$, $r > 0$, $\Lambda \geq 0$ and $\lambda > 0$ if there exist a linear function ℓ_H and a quadratic form q_H on the hyperplane $H := \xi^{\perp}$, with $q_H(h) \geq \lambda|h|^2$ for every $h \in H$, such that

$$\big|\mu(\xi + h) - \mu(\xi) - \ell_H(h) - q_H(h)\big| \leq \Lambda|h|^{2+\delta} \qquad \text{for every } h \in H \text{ with } |h| \leq r.$$

Theorem 1.1. *Let $d \geq 2$, and let ν satisfy* (W)*. For all $\delta \in (0,1]$, $r > 0$, $\Lambda \geq 0$ and $\lambda > 0$, there exist $\varepsilon > 0$ and $C < \infty$ such that, for every unit vector ξ satisfying* (Curv) *with these constants and every $x \in \mathbb{Z}^d \setminus \{0\}$ within distance $|x|^{3/4}$ of the ray $\mathbb{R}_+\xi$,*

$$\mathbb{E}\Big[\exp\Big(\frac{|T(0,x) - \mu(x)|}{C|x|^{\frac{1}{2}(1-\varepsilon)}}\Big)\Big] \leq 2\,.$$

Theorem 1.1 implies that $\mathrm{Var}[T(0,x)] \leq 2C^2|x|^{1-\varepsilon}$ for every such x. This answers a question of Kesten [Kes93, Remark 1] in every direction that satisfies (Curv); we refer to [AB26, Section 1] for the history of this question. No direction is known to satisfy (Curv) for any law of the weights. For exponential weights on $\mathbb{Z}^2$, strict convexity and differentiability of the limit shape were recently announced [Ope26]; (Curv) asks in addition for a second-order expansion with a nondegenerate quadratic form.

This paper is a companion to [AB26], where we prove the same bound for rotationally invariant first-passage percolation on $\mathbb{R}^d$. Both papers modify the induction on scales of Basu, Sidoravicius and Sly [BSS23, BSS26]. The two proofs could be written as one, under a set of conditions satisfied

by both models, but we found that this made the argument hard to follow. Instead, we assume that the reader has read [AB26].

Section 2 gives complete proofs of the results that replace isotropy and the lower bound on the density in [AB26], and Section 3 sets up the induction. Sections 4–7 follow [AB26, Sections 4–7]: each states the lattice versions of our results there and explains how the proofs change.

1.1. **Proof overview.** We describe how the proof overview of [AB26, Section 1.2] changes on the lattice. The notation is that of Section 1.2 below. We order the coordinates so that the first component ξ_1 of ξ has the largest absolute value, and we take it positive. We write points of $\mathbb{R}^d$ as (s,y) with $s \in \mathbb{R}$ and $y \in \mathbb{R}^{d-1}$, so that $\xi = \xi_1(1,t_0)$, and we call y/s the slope of (s,y). Fix a large integer K and a large constant A, and let $Q_s := As^{1/2-\delta/K}$. We induct on the following statement, for integers $s \geq 1$:

$$\mathbb{E}\big[\exp\big(|T(0,(s,y)) - \mu(s,y)|/Q_s\big)\big] \leq 2 \qquad \text{for } y \in \mathbb{Z}^{d-1} \text{ with } |y - st_0| \leq s^{3/4+\delta/100}.$$

It gives Theorem 1.1 with $\varepsilon = 2\delta/K$. Since it controls only the slopes within $s^{\delta/100-1/4}$ of t_0, most of our lemmas assume a bound on the slopes of the vectors they use.

The induction step derives the statement at $s = N$ from the statement at the integers $s \leq 2N/3$. As in [AB26], for $y \in \mathbb{Z}^{d-1}$ with $|y - Nt_0| \leq N^{3/4+\delta/100}$ and $T := T(0,(N,y))$, the statement at $s = N$ follows from $\mathbb{E}[T] \geq \mu(N,y)$ and the bounds

$$\text{(a)}\ \ \mathbb{E}\big[\exp\big(8|T - \mathbb{E}[T]|/Q_N\big)\big] \leq C\,, \qquad \text{(b)}\ \ \mathrm{Var}[T] \leq \omega_K Q_N^2\,, \qquad \text{(c)}\ \ \mathbb{E}[T] - \mu(N,y) \leq Q_N/8\,,$$

where C does not depend on A or K, and $\omega_K \to 0$ as $K \to \infty$.

To prove them, we compare T with passage times across the K slabs $[(i-1)n, in] \times \mathbb{R}^{d-1}$, where we assume that $n := N/K$ is an integer (Section 6 reduces to this case). On the lattice, a *slab* is a set of edges: for integers $p \leq q-2$, the slab $P = [p,q] \times \mathbb{R}^{d-1}$ consists of the edges $\{x, x+e_1\}$ with $p \leq x \cdot e_1 < q$ and $\{x, x+e_j\}$ with $p < x \cdot e_1 < q$ and $2 \leq j \leq d$, and its *faces* are the hyperplanes $\{x \cdot e_1 = p\}$ and $\{x \cdot e_1 = q\}$. Consecutive slabs have disjoint sets of edges, so the passage times $T^{(i)}$ along paths with edges in the ith slab are independent. Let $t := y/N$, and let

$$F := \min \sum_{i=1}^{K} T^{(i)}(x_{i-1}, x_i)\,,$$

where the minimum is over the *chains* of lattice points $0 = x_0, x_1, \ldots, x_K = (N,y)$ with $x_i \cdot e_1 = in$ and $|x_i - x_{i-1} - n(1,t)| \leq n^{3/4+\delta/100}/16$. A *minimizing chain* is one that attains the minimum. We show that $\mathbb{E}[(F-T)^2]^{1/2} \leq CK\log^2 N$.

This comparison, proved in Sections 2 and 6, needs the geodesics between the faces of a slab to stay in the slab. Since weights may vanish, we use a bound of Kesten instead of the lower bound on the density: for some $c > 0$ and every integer $m \geq 1$, with probability at least $1 - Ce^{-cm}$, every self-avoiding path from 0 with $m' \geq m$ edges has passage time greater than cm'. From this bound and (Curv), we prove the following bound on the directions of geodesics. With probability at least $1 - Ce^{-c\log^2 N}$, for all lattice points $u, v \in [-N^2, N^2]^d$ with $v - u = (\ell, \ell t')$, $1 \leq \ell \leq 3N$ and $|t' - t_0| \leq 3N^{-1/9}$, every geodesic from u to v satisfies

$$\angle(v-z, v-u) \leq C(\log N)^{1/2}|v-z|^{-1/4} + C|t'-t_0|^{1+\delta/2}\log N$$

at its vertices z with $|v - z| \geq C\log^2 N$.

Step 1 (Section 4). As in [AB26], we prove (a) and a bound uniform over two cubes: for an integer $s \in [N^{1/2}, 2N]$, a slope t' with $|t' - t_0| \leq s^{\delta/100-1/4}/2$, and two cubes of side $(sQ_s)^{1/2}$ in the hyperplanes $\{x \cdot e_1 = 0\}$ and $\{x \cdot e_1 = s\}$ whose centers differ by (s, st'),

$$\mathbb{E}\Big[\exp\Big(\sup_{u,v} |T(u,v) - \mu(v-u)|/(CQ_s)\Big)\Big] \leq 2\,.$$

As there, the bound follows from the statement by chaining over dyadic scales for $s \le N/2$, and from a union bound over chains of a fixed number of slabs for larger s; the latter also gives (a).

In [AB26], moving an endpoint of $[0, se_1]$ by $z \perp e_1$ lengthens the segment by about $|z|^2/(2s)$. On the lattice, μ changes to first order in z, and we subtract this change with a subgradient $\zeta_{t'}$ of μ at $(1,t')$, that is, a vector with $\zeta_{t'} \cdot (1,t') = \mu(1,t')$ and $\zeta_{t'} \cdot x \le \mu(x)$ for every $x \in \mathbb{R}^d$. By (Curv), for $z \in \mathbb{R}^{d-1}$ with $|z| \le cs$, and $|t' - t_0| \le c$,

$$c\frac{|z|^2}{s} - Cs|t'-t_0|^{2+\delta} \le \mu(s, st'+z) - \zeta_{t'} \cdot (s, st'+z) \le C\frac{|z|^2}{s} + Cs|t'-t_0|^{2+\delta},$$

and $s|t'-t_0|^{2+\delta}$ is much smaller than Q_s. Along a chain, the terms $\zeta_{t'} \cdot (x_i - x_{i-1})$ add up to $\zeta_{t'} \cdot (x_K - x_0)$, which does not depend on the chain. So the arguments of [AB26] apply with $\zeta_{t'} \cdot (v-u)$ in place of $(v-u) \cdot e_1$.

Step 2 (Section 5). As in [AB26], for an integer $\ell \in [N/(9K), N]$ and two cubes as in Step 1 with $s = \ell$, the geodesics between the cubes stay within distance of order $(\ell Q_\ell)^{1/2}$ of the segments between their endpoints, and $T(u,v) - \zeta_{t'} \cdot (v-u)$ varies by at most ηQ_ℓ in mean square when u and v move by $h(\ell Q_\ell)^{1/2}$, with h small depending on η. We obtain moments of every order instead of Gaussian tails: for every $p \ge 1$,

$$\mathbb{E}\Big[\sup_{u,v,\pi,x} \operatorname{dist}(x,[u,v])^p\Big] \le C(p)(\ell Q_\ell)^{p/2},$$

where u and v range over the two cubes, π over the geodesics from u to v in the slab between the cubes, and x over the vertices of π. The bound on the directions of geodesics replaces the confinement to a box by the lower bound on the density: it keeps them within distance $\ell^{3/4+\delta/200}$ of the line through the centers of the cubes.

Step 3 (Section 6). We prove (b) as in [AB26], with the inequality of Benaïm and Rossignol [BR08, Proposition 2.1] and the averaging argument of Benjamini, Kalai and Schramm [BKS03, Lemma 3]. Three things change. First, the blocks follow the slope t: with $w := \lceil (nQ_n)^{1/2} \rceil$ and a large integer L, they are the sets of edges of the ith slab whose midpoints (q,h) satisfy $h - qt \in w(z + [0,L)^{d-1})$, for $1 \le i \le K$ and $z \in L\mathbb{Z}^{d-1}$. Second, in place of the passage time across a block, the variable X_G of a block G discretizes the conditional expectation, given the weights in G, of the normalized slab passage times $(T^{(i)}(u,v) - \zeta_t \cdot (v-u))/Q_n$ across G. Third, F is averaged over its translates by the lattice vectors $mLwe_2$, for $0 \le m < \lceil K^{2/3} \rceil$.

Step 4 (Section 7). As in [AB26], we prove (c) by showing that $\mathbb{E}[T(0,(s,y_0))] - \mu(s,y_0) \le Q_s/100$ for $s := \lfloor N/4 \rfloor$ and a lattice point y_0 within distance $2(d-1)^{1/2}$ of sy/N. If this fails, [AB26] bounds the means of the slab passage times from below by isotropy. Instead, let ζ be a subgradient of μ at $(1, y_0/s)$. The continuity of Step 2, for $|z| \le h(sQ_s)^{1/2}$ with h small, and, for larger $|z|$, the bound $\mathbb{E}[T] \ge \mu$ with the lower bound of Step 1, give

$$\mathbb{E}[T(0,(s,y_0+z))] - \zeta \cdot (s, y_0+z) \ge c\big(Q_s + |z|^2/s\big) \qquad \text{for } z \in \mathbb{Z}^{d-1} \text{ with } |z| \le s^{3/4+\delta/100}/4.$$

Let F be the minimum defined as above, for s slabs of width s from 0 to (s^2, sy_0). By (b) for the passage times to the points (s,y), the slab passage times between cubes of side $(sQ_s)^{1/2}$ fluctuate by at most a small multiple of Q_s, so, as in [AB26], summing this bound along a minimizing chain gives $\mathbb{E}[F] - s\mu(s,y_0) \ge csQ_s$. This contradicts the upper bound $\mathbb{E}[F] - s\mu(s,y_0) \le Cs\log^2 s$, as in [AB26].

1.2. **Notation and conventions.** We use the notation of [AB26, Section 1.3] and [AKM19, (0.41)–(0.42)] for exponential bounds: for $s, \theta > 0$, a random variable X satisfies $X \le \mathcal{O}_s(\theta)$ if $\mathbb{E}[\exp((X_+/\theta)^s)] \le 2$, where $X_+ := \max\{X, 0\}$, and $X \le Y + \mathcal{O}_s(\theta)$ means $X - Y \le \mathcal{O}_s(\theta)$. If $X_j \le \mathcal{O}_s(\theta)$ for $j \le J$, then

$$\max_{j \le J} X_j \le \mathcal{O}_s\big(C(s)\theta \log^{1/s}(2+J)\big). \tag{1.1}$$

If $s \geq 1$ and $X_j \leq \mathcal{O}_s(\theta_j)$ for every j, then, by [AKM19, Lemma A.4] applied to the positive parts,

$$\sum_j X_j \leq \mathcal{O}_s\Big(\sum_j \theta_j\Big). \tag{1.2}$$

Constants and thresholds depend on d, ν and the constants $\delta, r, \Lambda, \lambda$ of (Curv), but not on ξ. Any other dependence is written out: for instance, $C(L)$ depends also on L. The constants $c, C > 0$ may change from line to line. We write $[u, v]$ for the segment from u to v, and $\angle(x, y) \in [0, \pi]$ for the angle between nonzero vectors x and y. We write $e_1, \dots, e_d$ for the standard basis of $\mathbb{R}^d$, $|x|_1 := \sum_{j=1}^d |x \cdot e_j|$ for $x \in \mathbb{R}^d$ and $\mathbf{1}_E$ for the indicator of an event E.

1.3. **Formalization.** All the results of this paper are formalized in the Lean 4 proof assistant, with the Mathlib library, in the repository

https://github.com/nitromannitol/FPP-variance

It also contains the companion paper [AB26] on rotationally invariant first-passage percolation, with the formalization of all its results, and checked lists of the errors and gaps found in [BSS23] and [BSS26].

1.4. **Use of artificial intelligence.** This paper and [AB26] grew out of the project described in [AB26, Section 1.5]. The induction on scales of both papers comes from the work of Basu, Sidoravicius and Sly [BSS23, BSS26], and the present paper would not exist without their pioneering work.

A variance bound of the form of Theorem 1.1 was first proved by GPT-6 Astra and GPT-6.1 Sol agents, under (Curv), on $\mathbb{Z}^2$ in the direction e_1, and then in every dimension and every direction that satisfies (Curv). An initial draft of the present paper was written by Claude Opus 5.5 under our direction, and the authors subsequently heavily revised and rewrote this draft. The formalization described in Section 1.3 was written with the help of AI agents.

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2. Preliminaries

This section proves the results that replace isotropy and the bound on the density in [AB26]: two consequences of (Curv), classical bounds on passage times and geodesics, and the comparison of slab passage times with T.

Throughout, ν satisfies (W) and ξ satisfies (Curv). Since permutations and sign changes of the coordinates preserve the law of the weights, μ and (Curv), we order and orient the coordinates so that the first component ξ_1 of ξ is positive and has the largest absolute value. We write points of $\mathbb{R}^d$ as (s, y) with $s \in \mathbb{R}$ and $y \in \mathbb{R}^{d-1}$, and $e_1, \dots, e_d$ for the resulting basis, so that $\xi = \xi_1(1, t_0)$ with $\xi_1 \geq d^{-1/2}$ and $|t_0| \leq (d-1)^{1/2}$.

When a point $x \in \mathbb{R}^d$ is used as an endpoint of a passage time, we replace it by the point of $\mathbb{Z}^d$ with the coordinates $\lceil x \cdot e_j - 1/2 \rceil$, a nearest point of $\mathbb{Z}^d$ to x. Let μ_* and μ^* be the minimum and the maximum of μ on the unit sphere, so that $\mu_*|x| \leq \mu(x) \leq \mu^*|x|$ for every $x \in \mathbb{R}^d$.

For integers p and q with $q - p \geq 2$, the *slab* $P = [p, q] \times \mathbb{R}^{d-1}$ consists of the edges $\{x, x + e_1\}$ with $p \leq x \cdot e_1 < q$ and $\{x, x + e_j\}$ with $p < x \cdot e_1 < q$ and $2 \leq j \leq d$. The edges in its *faces* $\{x \cdot e_1 = p\}$ and $\{x \cdot e_1 = q\}$ are omitted, so consecutive slabs have disjoint sets of edges.

For lattice points u and v of $[p, q] \times \mathbb{R}^{d-1}$, let $T_P(u, v) := \inf_\pi \sum_{e \in \pi} \tau_e$, where the infimum is over the paths π from u to v with edges in P. A *geodesic* is a self-avoiding path that attains the passage time between its endpoints, and a geodesic in P is one with edges in P that attains T_P.

Since the faces of the slabs are orthogonal to e_1, we expand μ on $\{x \cdot e_1 = 1\}$ rather than on $\xi + H$.

Lemma 2.1. *Let $r_0 := \min\{1, r\}/4$. There exist an affine function ψ on $\mathbb{R}^{d-1}$ and a symmetric matrix Σ, with $c|h|^2 \le h \cdot \Sigma h \le C|h|^2$ for every $h \in \mathbb{R}^{d-1}$, such that*

$$\big|\mu(1, t_0 + h) - \psi(h) - h \cdot \Sigma h\big| \le C|h|^{2+\delta} \qquad \text{for every } h \in \mathbb{R}^{d-1} \text{ with } |h| \le r_0. \tag{2.1}$$

Proof. For $h \in \mathbb{R}^{d-1}$, let $v_h := (0, h) - (\xi \cdot (0, h))\xi$ be the orthogonal projection of $(0, h)$ onto H, and let $\rho_h := 1/\xi_1 + \xi \cdot (0, h)$, so that $(1, t_0 + h) = \rho_h \xi + v_h$. We have $|v_h| \le |h|$ and $|v_h|^2 = |h|^2 - (\xi \cdot (0, h))^2 \ge \xi_1^2 |h|^2$, since the last $d-1$ components of ξ form the vector $\xi_1 t_0$, of length $(1 - \xi_1^2)^{1/2}$. If $|h| \le r_0 \le 1/(4\xi_1)$, then $1/(2\xi_1) \le \rho_h \le 3/(2\xi_1)$ and $|v_h|/\rho_h \le 2\xi_1 |h| \le r$. Hence homogeneity and (Curv) give

$$\mu(1, t_0 + h) = \rho_h \mu(\xi) + \ell_H(v_h) + \rho_h^{-1} q_H(v_h) + E_h\,, \qquad |E_h| \le \rho_h \Lambda\big(|v_h|/\rho_h\big)^{2+\delta} \le 12\Lambda |h|^{2+\delta}\,,$$

since $\rho_h (2\xi_1)^{2+\delta} \le 3(2\xi_1)^{1+\delta} \le 12$. Let $\psi(h) := \rho_h \mu(\xi) + \ell_H(v_h)$, which is affine in h, and let Σ be the symmetric matrix with $h \cdot \Sigma h = \xi_1\, q_H(v_h)$.

Adding the inequality of (Curv) at w and at $-w$, for $w \in H$ with $|w| = r$, and using $\mu(\xi \pm w) \le \mu(\xi) + \mu^* r$, gives $q_H(w) \le \mu^* r + \Lambda r^{2+\delta}$, so $q_H(w) \le C|w|^2$ for every $w \in H$ by homogeneity. Hence

$$\lambda d^{-3/2} |h|^2 \le \lambda \xi_1^3 |h|^2 \le h \cdot \Sigma h \le C|h|^2\,.$$

Finally,

$$|\rho_h^{-1} - \xi_1| = \xi_1 |\xi \cdot (0, h)|/\rho_h \le 2\xi_1^2 |h|\,,$$

so replacing ρ_h^{-1} by ξ_1 changes the third term by at most $C|h|^3 \le C|h|^{2+\delta}$. □

The *slope* of a vector (s, y) with $s \ne 0$ is y/s. For $t \in \mathbb{R}^{d-1}$, since μ is convex and positively homogeneous, a vector ζ_t is a *subgradient* of μ at $(1, t)$ if and only if $\zeta_t \cdot (1, t) = \mu(1, t)$ and $\mu(z) \ge \zeta_t \cdot z$ for every $z \in \mathbb{R}^d$. Subgradients exist since μ is a norm, and every one satisfies $|\zeta_t| \le \mu^*$, since $|\zeta_t|^2 \le \mu(\zeta_t) \le \mu^* |\zeta_t|$. The next lemma bounds $\mu(z) - \zeta_t \cdot z$ from both sides at $z = (1, t + h)$, and from below at every z.

Lemma 2.2. *There exists $r_1 \in (0, r_0/2]$ such that, for all $t, h \in \mathbb{R}^{d-1}$ with $|t - t_0| \le r_1$ and $|h| \le r_1$, every subgradient ζ_t of μ at $(1, t)$ and every $z \in \mathbb{R}^d \setminus \{0\}$,*

$$c|h|^2 - C|t - t_0|^{2+\delta} \le \mu(1, t + h) - \zeta_t \cdot (1, t + h) \le C|h|^2 + C|t - t_0|^{2+\delta}\,, \tag{2.2}$$

$$\mu(z) - \zeta_t \cdot z \ge c|z| \angle(z, (1, t))^2 - C|z||t - t_0|^{2+\delta}\,. \tag{2.3}$$

Proof. Since

$$\zeta_t \cdot (1, t + h) = 2\zeta_t \cdot (1, t + h/2) - \zeta_t \cdot (1, t) = 2\zeta_t \cdot (1, t) - \zeta_t \cdot (1, t - h)\,,$$

the middle term of (2.2) lies between the second differences

$$\mu(1, t + h) - 2\mu(1, t + h/2) + \mu(1, t) \qquad \text{and} \qquad \mu(1, t + h) + \mu(1, t - h) - 2\mu(1, t)\,.$$

By (2.1), the affine parts of these second differences cancel, their quadratic parts are $h \cdot \Sigma h/2$ and $2h \cdot \Sigma h$, and their remainders are at most $C(|h|^{2+\delta} + |t - t_0|^{2+\delta})$. Taking r_1 small proves (2.2).

For (2.3), let $\phi(z) := \mu(z) - \zeta_t \cdot z$, which is nonnegative, positively homogeneous, subadditive and $2\mu^*$-Lipschitz, with $\phi(1, t) = 0$. By homogeneity, we may assume that $|z| = 1$. We write $z = (z_1, z')$, with $z' \in \mathbb{R}^{d-1}$, and let $\hat{e} := (1, t)/|(1, t)|$ and $\theta := \angle(z, \hat{e})$. Let $\beta := \min\{1/2, r_1/(2(1 + (d-1)^{1/2} + r_1))\}$, so that $2\beta(1 + |t|) \le r_1$. With $\sigma := 1 + \beta z_1 \in [1/2, 3/2]$,

$$(1, t) + \beta z = \sigma(1, t + h)\,, \qquad h := \frac{\beta}{\sigma}(z' - z_1 t)\,, \qquad |h| \le 2\beta(1 + |t|) \le r_1\,.$$

Subadditivity, homogeneity and $\phi(1, t) = 0$ give $\sigma\phi(1, t + h) \le \beta\phi(z)$, so (2.2) gives

$$\phi(z) \ge \frac{\sigma}{\beta}\big(c|h|^2 - C|t - t_0|^{2+\delta}\big) \ge \frac{c\beta}{2}|z' - z_1 t|^2 - \frac{2C}{\beta}|t - t_0|^{2+\delta}\,.$$

Since $(0, z' - z_1 t) = z - z_1(1,t)$, we have

$$|z' - z_1 t|^2 \geq \sin^2\theta = (1-\cos\theta)|z+\hat{e}|^2/2 \geq \pi^{-2}\theta^2|z+\hat{e}|^2 .$$

If $|z+\hat{e}| \geq \min\{1, \mu_*/(2\mu^*)\}$, this gives (2.3). Otherwise, since $\phi(-\hat{e}) = 2\mu(\hat{e}) \geq 2\mu_*$, the Lipschitz bound gives

$$\phi(z) \geq 2\mu_* - 2\mu^*|z+\hat{e}| \geq \mu_* \geq \mu_*\theta^2/\pi^2 .$$

□

To round points of $\mathbb{R}^{d-1}$ to $\mathbb{Z}^{d-1}$ within a given ball, we use the following fact. If $x, x' \in \mathbb{R}^{d-1}$ and $R \geq 2(d-1)^{1/2}$ satisfy $|x - x'| \leq R$, then some $z \in \mathbb{Z}^{d-1}$ satisfies

$$|z - x'| \leq R \qquad \text{and} \qquad |z - x| \leq 2(d-1)^{1/2} , \tag{2.4}$$

namely a nearest lattice point to the point of $[x, x']$ at distance $\min\{|x-x'|, (d-1)^{1/2}\}$ from x.

We recall the classical bounds, the counterpart of [AB26, Lemma 2.1]. By (W) and (1.2), the passage time along a path with m edges is at most $\mathcal{O}_1(Cm)$. For $x \in \mathbb{Z}^d \setminus \{0\}$,

$$|T(0,x) - \mathbb{E}[T(0,x)]| \leq \mathcal{O}_1(C|x|_1^{1/2}) \qquad \text{and} \qquad 0 \leq \mathbb{E}[T(0,x)] - \mu(x) \leq C(|x|_1 \log(2+|x|_1))^{1/2} . \tag{2.5}$$

The first bound follows from [ADH17, Theorem 3.13], which gives Gaussian tails for the lower deviations at all scales, and for the upper deviations up to a fixed multiple of $|x|_1$. The Chernoff bound for the weights along a path of $|x|_1$ edges gives the larger upper deviations. The second bound is [ADH17, Theorem 3.33] for $|x|_1 \geq 2$. For $|x|_1 = 1$, it follows from $\mu(x) \leq \mathbb{E}[T(0,x)] \leq \mathbb{E}[\tau_e]$.

Kesten's bound on paths of small passage time [Kes86, Proposition 5.8] assumes that $\nu(\{0\})$ is less than the critical probability $p_T(d)$ for the finiteness of the expected cluster size of bond percolation. By the sharpness of the phase transition [DCT16, Theorem 1.1], $p_T(d) = p_c(d)$, so it applies under (W). It gives $c > 0$ such that, for every $x \in \mathbb{Z}^d$, the largest m for which some self-avoiding path from x with m edges has passage time at most cm is at most $\mathcal{O}_1(C)$.

Minimizing paths exist [ADH17, Proposition 4.4], and removing their loops gives geodesics. By Kesten's bound and the Chernoff bound for the weights along a path from u to v with $|v-u|_1$ edges, every geodesic from u to v has at most $C(1+|v-u|_1) + \mathcal{O}_1(C)$ edges, for all $u, v \in \mathbb{Z}^d$.

Geodesics in a slab $P = [p,q] \times \mathbb{R}^{d-1}$ also exist. Two lattice points u, v of $[p,q] \times \mathbb{R}^{d-1}$ are joined by a path in P with at most $|v-u|_1 + 2$ edges, so $T_P(u,v) < \infty$. By Kesten's bound, only finitely many self-avoiding paths from u have passage time at most $T_P(u,v)+1$. Hence geodesics in P exist, and they satisfy the same bound on the number of edges.

The next lemma collects these bounds on one event, uniformly over polynomially many endpoints.

Lemma 2.3. *For every sufficiently large integer N and every $S \geq \log N$, there exists an event $\mathcal{G}_N(S)$, increasing in S, with $\mathbf{1}_{\mathcal{G}_N(S)^c} \leq \mathcal{O}_2(C/S)$, on which the following hold for all $x, y \in [-2N^2, 2N^2]^d \cap \mathbb{Z}^d$:*

1. *$|T(x,y) - \mu(y-x)| \leq C(S|y-x|^{1/2} + S^2)$;*
2. *$T_P(x,y) \leq C(|y-x| + S^2)$ for every slab $P = [p,q] \times \mathbb{R}^{d-1}$ with $x, y \in [p,q] \times \mathbb{R}^{d-1}$;*
3. *every self-avoiding path from x with $m \geq CS^2$ edges has passage time at least cm.*

Proof. Let $\mathcal{G}_N(S)$ be the event on which (1)–(3) hold. The Gaussian tails of [ADH17, Theorem 3.13], the Chernoff bound for the weights along a path of $|y-x|_1$ edges, and the mean bound in (2.5) show that each instance of (1) fails with probability at most Ce^{-cS^2}. Here we use that $S \geq \log N \geq C(\log(2+|y-x|_1))^{1/2}$ for N large.

The Chernoff bound for a path in P from x to y with at most $|y-x|_1 + 2$ edges gives the same for (2), and Kesten's bound gives it for (3). Since $T_P(x,y)$ does not increase when $[p,q]$ is enlarged, it suffices to prove (2) when $\min\{x \cdot e_1, y \cdot e_1\} - 2 \leq p$ and $q \leq \max\{x \cdot e_1, y \cdot e_1\} + 2$.

A union bound over the at most CN^C choices of x, y, p and q shows that (1)–(3) hold off an event of probability at most $CN^C e^{-cS^2} \le e^{-cS^2/2}$, since $S \ge \log N$ and N is large. This gives $\mathbf{1}_{\mathcal{G}_N(S)^c} \le \mathcal{O}_2(C/S)$. The event $\mathcal{G}_N(S)$ increases with S, since the three bounds weaken as S increases. □

On $\mathcal{G}_N(S)$, the next lemma bounds the angles $\angle(v-z,(1,t))$ and $\angle(z-u,(1,t))$ at the vertices z of the paths from u to v with passage time at most $T(u,v)+S^2$, where t is the slope of $v-u$. It replaces [AB26, Lemma 2.2].

Lemma 2.4. *Let N and S be as in Lemma 2.3, and let $u, v \in [-N^2, N^2]^d \cap \mathbb{Z}^d$ with $v-u=(\ell,\ell t)$, $1 \le \ell \le 3N$ and $|t-t_0| \le 3N^{-1/9}$. If N is sufficiently large, then, on $\mathcal{G}_N(S)$, every self-avoiding path from u to v with passage time at most $T(u,v)+S^2$ satisfies, at its vertices z with $|v-z| \ge CS^2$,*

$$\angle\big(v-z,(1,t)\big) \le C\big(S^{1/2}|v-z|^{-1/4} + |t-t_0|^{1+\delta/2}\log N\big)\,, \tag{2.6}$$

and the same bound, with $|z-u|$ in place of $|v-z|$, holds for $\angle(z-u,(1,t))$ at its vertices with $|z-u| \ge CS^2$.

Proof. By Lemma 2.3, such a path has at most $C(N+S^2)$ edges, and each of its subpaths, from x to y, has passage time at most $T(x,y)+S^2$. If $S^2 \ge N$, then $|v-z| \le C(N+S^2) \le 2CS^2$ for every vertex z, so the lemma is vacuous once the constant in $|v-z| \ge CS^2$ exceeds $2C$. Otherwise, the path stays in $[-2N^2, 2N^2]^d$. For every $R \ge S^2$, Lemma 2.3(1) gives

$$(*) \qquad \mu(x'-x) + \mu(x''-x') \le \mu(x''-x) + CSR^{1/2}$$

for all its vertices x, x', x'', in this order or in the reverse order, with $|x'-x|, |x''-x'| \le R$. Inequality $(*)$ implies the following, for $R \ge CS^2$.

- $(**)$ The path has no vertices x, x', x'', in this order or in the reverse order, with $\mu(x'-x) \ge R$ and $\mu(x''-x) \le R/2$.

To prove $(**)$, suppose that x, x', x'' are such vertices, and let $\tilde{x}$ be the first vertex after x, along the path from x towards x'', with $\mu(\tilde{x}-x) \ge R$. Then $\tilde{x}$ lies between x and x'', and $\mu(\tilde{x}-x) \le R+\mu^*$, so $|\tilde{x}-x|, |x''-\tilde{x}| \le CR$. Since $\mu(x''-\tilde{x}) \ge \mu(\tilde{x}-x) - \mu(x''-x) \ge R/2$, inequality $(*)$ for $x, \tilde{x}, x''$, with CR in place of R, gives $3R/2 \le R/2 + CSR^{1/2}$, which is false.

Let $R_j := 2^{-j}\mu(v-u)$ for $j \ge 0$. Let $z_0 := u$ and, for $j \ge 1$, let z_j be the last vertex of the path with $\mu(v-z_j) \ge R_j$, so that $R_j \le \mu(v-z_j) \le R_j + \mu^*$. Let $\theta_j := \angle(v-z_j,(1,t))$, and let t_j' be the slope of $v-z_j$. For large constants C_2 and C_3, we show by induction on j that

$$\theta_j \le C_2\big(S^{1/2}R_j^{-1/4} + j|t-t_0|^{1+\delta/2}\big) \qquad \text{if } R_j \ge C_3S^2. \tag{2.7}$$

It holds at $j=0$, since $\theta_0 = 0$.

Suppose that it holds at some j with $R_j \ge C_3S^2$. Since $j \le C\log N$ and $|t-t_0| \le 3N^{-1/9}$, it gives $\theta_j \le C_2(C_3^{-1/4} + CN^{-1/9}\log N)$. If C_3 is large depending on C_2 and N is large, this is at most $\min\{1, r_1\}/C$, so $|t_j'-t| \le C\theta_j$ and $|t_j'-t_0| \le r_1$. Let ζ be a subgradient of μ at $(1,t_j')$, so that $\zeta\cdot(v-z_j) = \mu(v-z_j)$.

By $(**)$ and the choice of z_j, the vertices z between z_j and z_{j+1} satisfy $R_j/4 \le \mu(v-z) \le 2R_j + \mu^*$; for $j=0$, the upper bound is $(**)$ for the vertices v, z, u, in the reverse order, with $R := 2R_0$. For these z, inequality $(*)$, applied to z_j, z, v with CR_j in place of R, gives

$$\mu(v-z) - \zeta\cdot(v-z) \le \mu(z-z_j) + \mu(v-z) - \mu(v-z_j) \le CSR_j^{1/2}\,.$$

Hence (2.3), with t_j' in place of t, gives

$$\angle\big(v-z,(1,t)\big) \le \theta_j + \angle\big(v-z,(1,t_j')\big) \le \theta_j + CS^{1/2}R_j^{-1/4} + C\big(|t-t_0| + C\theta_j\big)^{1+\delta/2}\,.$$

By (2.7) at j, the last term is at most $C_4(|t-t_0|^{1+\delta/2} + S^{1/2}R_j^{-1/4})$, with C_4 independent of C_2, if C_3 is large depending on C_2 and N is large. Indeed, $(x+y)^{1+\delta/2} \le 2(x^{1+\delta/2} + y^{1+\delta/2})$ for $x, y \ge 0$, and

$$\left(C_2 S^{1/2} R_j^{-1/4}\right)^{1+\delta/2} \le S^{1/2} R_j^{-1/4} \qquad \text{and} \qquad \left(C_2 j |t-t_0|^{1+\delta/2}\right)^{1+\delta/2} \le |t-t_0|^{1+\delta/2},$$

the first since $R_j \ge C_3 S^2$, and the second since $j \le C \log N$ and $|t-t_0| \le 3N^{-1/9}$. With $z = z_{j+1}$ and $R_{j+1} = R_j/2$, this gives (2.7) at $j+1$ if $(2^{1/4}-1)C_2 \ge C + C_4$ and $C_2 \ge C_4$. For every vertex z between z_j and z_{j+1}, it gives (2.6), since $|v-z| \le CR_j$ and $j \le C \log N$.

The remaining vertices z satisfy $|v-z| \le CS^2$. If $R_0 \ge C_3 S^2$, they come after z_{J+1}, where J is the largest j with $R_j \ge C_3 S^2$, so $\mu(v-z) < R_{J+1} < C_3 S^2$. Otherwise, $(**)$ for the vertices v, z, u, in the reverse order, with $R := \max\{2\mu(v-u), CS^2\}$, gives $\mu(v-z) < R \le CS^2$. Finally, since $\mu(-z) = \mu(z)$, inequality $(*)$ holds also for the path with the vertices $u+v-z$, traversed from u to v. The argument above uses only this inequality, the unit steps of the path and the bound on its number of edges, which hold also for this path. Hence it applies to this path, and $v-(u+v-z) = z-u$ gives the bound for $\angle(z-u,(1,t))$. □

Let $M \ge 1$ and $n \ge 64d$ be integers, and let $u, v \in \mathbb{Z}^d$ with $v-u = (Mn, Mnt)$ for some $t \in \mathbb{R}^{d-1}$. For $1 \le i \le M$, let $T^{(i)}$ be the T_P of the ith slab $P = [u\cdot e_1 + (i-1)n, u\cdot e_1 + in] \times \mathbb{R}^{d-1}$, and let

$$F := \min \sum_{i=1}^{M} T^{(i)}(x_{i-1}, x_i)\,, \tag{2.8}$$

where the minimum is over the *chains* $u = x_0, x_1, \dots, x_M = v$ of points of $\mathbb{Z}^d$ with $x_i \cdot e_1 = u\cdot e_1 + in$ and $|x_i - x_{i-1} - n(1,t)| \le n^{3/4+\delta/100}/16$. Chains exist, since the points of $\mathbb{Z}^d$ to which the points $u + in(1,t)$ are rounded form one. A *minimizing chain* is one that attains the minimum. Since the slabs have disjoint sets of edges, the $T^{(i)}$ are independent.

On $\mathcal{G}_N(S)$, the next lemma compares slab passage times with T, and F with $T(u,v)$, as in [AB26, Lemma 2.3].

Lemma 2.5. *Let N and S be as in Lemma 2.3. If N is sufficiently large, then the following hold for all $u, v \in \mathbb{Z}^d$ with $v-u = (\ell, \ell t)$, $1 \le \ell \le 3N$ and $|t-t_0| \le 3N^{-1/9}$.*

1. *If $u, v \in [-N^2, N^2]^d$ and $\ell \ge CS^2$, then, on $\mathcal{G}_N(S)$, $0 \le T_P(u,v) - T(u,v) \le CS^2$ for the slab $P := [u\cdot e_1, v\cdot e_1] \times \mathbb{R}^{d-1}$.*
2. *If $\ell = Mn$ for integers $M \ge 1$ and $n \ge N^{1/2}$, $|t-t_0| \le 2n^{\delta/100 - 1/4}$, $S \le N^{\delta/1000}$, and u and v lie in $[-3(1+|t_0|)N, 3(1+|t_0|)N]^d$, then, on $\mathcal{G}_N(S)$, $0 \le F - T(u,v) \le CMS^2$.*

Proof. The lower bounds follow from $T_P \ge T$ and the triangle inequality.

For (1), let C_1 be large. Since $|t_0| \le (d-1)^{1/2}$, the vectors at angle at most $1/(8d^{1/2})$ from $(1,t)$ have positive first coordinate.

Let x be the last vertex of a geodesic from u to v at distance less than C_1S^2 from u, and x' the first vertex after x at distance less than C_1S^2 from v. At the vertices z between x and x', the right side of (2.6) is at most $CC_1^{-1/4} + C|t-t_0|^{1+\delta/2}\log N$. This is less than $1/(8d^{1/2})$ if C_1 and N are large, so $z-u$ and $v-z$ have positive first coordinates. Hence $u\cdot e_1 < z\cdot e_1 < v\cdot e_1$ at these vertices, which exist since $\ell \ge CS^2 \ge 3C_1S^2$.

Every edge with such an endpoint is in P, so the part of the geodesic from x to x' has its edges in P. Joining this part to u and v by geodesics in P, whose passage times are at most CS^2 by Lemma 2.3(2), gives $T_P(u,v) \le T(u,v) + CS^2$.

For (2), fix a geodesic from u to v, and, for $0 < i < M$, let x_i be its first vertex on the hyperplane $\{x\cdot e_1 = u\cdot e_1 + in\}$. The geodesic has at most $C(N+S^2)$ edges, so it lies in $[-N^2, N^2]^d$. At distance at least n from its endpoints, the right side of (2.6) is much smaller than $n^{\delta/100-1/4}$ for N large, since $S^{1/2}n^{-\delta/100}$ and $n^{-\delta/2(1/4-\delta/100)}\log N$ tend to zero.

Let $x_0 := u$ and $x_M := v$. For $0 \le i < M$, write $v - x_i = (M-i)n(1,t_i')$. Then (2.6) at x_i shows that $|t_i' - t|$ is much smaller than $n^{\delta/100-1/4}$, so $|t_i' - t_0| \le 3n^{\delta/100-1/4} \le 3N^{-1/9}$. The part of the geodesic from x_i is a geodesic from x_i to v, and Lemma 2.4 with t_i' in place of t, applied to it at its vertex x_{i+1}, bounds $\angle(x_{i+1} - x_i, (1,t_i'))$. Since $x_{i+1}\cdot e_1 - x_i \cdot e_1 = n$, these two bounds give $|x_{i+1} - x_i - n(1,t)| \le n^{3/4+\delta/100}/16$ for N large. Hence the x_i form a chain.

The passage times of the pieces of the geodesic between consecutive x_i add up to $T(u,v)$, and, by (1), each piece has passage time at least $T^{(i)}(x_{i-1},x_i) - CS^2$. Hence $F \le T(u,v) + CMS^2$. □

3. The induction

As in [AB26, Section 3], we prove Theorem 1.1 by induction, here on the first coordinate s of the endpoint (s,y). For an integer $K \ge 10^4$ and $A \ge 1$, let $Q_s := As^{1/2-\delta/K}$ for $s > 0$; the exponent δ/K, in place of the $1/K$ of [AB26], makes the condition $K \ge 10^4$ independent of δ. For an integer $s \ge 1$, let $\mathcal{D}(s) := \{y \in \mathbb{Z}^{d-1} : |y - st_0| \le s^{3/4+\delta/100}\}$. For an integer $N \ge 2$, the *induction hypothesis* is

$$|T(0,(s,y)) - \mu(s,y)| \le \mathcal{O}_1(Q_s) \qquad \text{for all integers } 1 \le s \le 2N/3 \text{ and all } y \in \mathcal{D}(s), \tag{3.1}$$

and the conclusion of the induction step is

$$|T(0,(N,y)) - \mu(N,y)| \le \mathcal{O}_1(Q_N/2) \qquad \text{for all } y \in \mathcal{D}(N). \tag{3.2}$$

Proposition 3.1 (Induction step)**.** *There exist an integer $K \ge 10^4$ and $N_0 \ge 2$ such that (3.1) implies (3.2) for every $A \ge 1$ and every integer $N \ge N_0$.*

Proof of Theorem 1.1, given Proposition 3.1. Let K and N_0 be as in Proposition 3.1. For integers $1 \le s < N_0$ and $y \in \mathcal{D}(s)$, the passage time $T(0,(s,y))$ is at most the passage time along a path of at most CN_0 edges, so the bound in (3.1) holds for these s once A is large, depending on K. By Proposition 3.1 and induction on N, it holds for every integer $s \ge 1$.

Let $\varepsilon := 2\delta/K$, so that $Q_s = As^{\frac12(1-\varepsilon)}$. Let $x = (s,y) \in \mathbb{Z}^d \setminus \{0\}$ lie within distance $|x|^{3/4}$ of $\mathbb{R}_+\xi$, and let $\beta\xi$ be a nearest point of $\mathbb{R}_+\xi$ to x. Since $x - \beta\xi = (s - \beta\xi_1, y - \beta\xi_1 t_0)$, we have $s \ge \xi_1|x|/2$ and $|y - st_0| \le (1+|t_0|)|x|^{3/4}$ if $|x|$ is large, so that $y \in \mathcal{D}(s)$ and $Q_s \le A|x|^{\frac12(1-\varepsilon)}$. Hence the bound in (3.1) gives the theorem for these x, with A in place of C.

The other x satisfy $|x| \le C$, and for them $|T(0,x) - \mu(x)|$ is at most C plus the passage time along a path of $|x|_1$ edges, which has an exponential moment by (W). □

The rest of the paper proves Proposition 3.1. By (2.5), for $y \in \mathcal{D}(N)$,

$$|T(0,(N,y)) - \mu(N,y)| \le \mathcal{O}_1\big(C(N\log(2+N))^{1/2}\big)\,,$$

which gives (3.2) if $Q_N \ge 2C(N\log(2+N))^{1/2}$. We may therefore assume that, with this constant C,

$$Q_N \le 2C(N\log(2+N))^{1/2}\,. \tag{3.3}$$

From now on, we assume (3.1) and (3.3). Since (3.3) gives $A \le CN^{\delta/K}(\log(2+N))^{1/2}$ and $K \ge 10^4$, we have, for $s \ge N^{1/2}/16$ and $t \in \mathbb{R}^{d-1}$ with $|t - t_0| \le 3s^{\delta/100-1/4}$,

$$\frac{(sQ_s)^{1/2}}{s^{3/4+\delta/100}} \le CN^{-\delta/250}(\log(2+N))^{1/4} \qquad \text{and} \qquad \frac{s|t-t_0|^{2+\delta}}{Q_s} \le Cs^{-\delta/5}\,. \tag{3.4}$$

As in [AB26, Section 1.2], for every integer $\ell \in [N/8, N]$, $y \in \mathcal{D}(\ell)$ and $T := T(0,(\ell,y))$, we prove the bounds

$$\text{(a)}\ \ \mathbb{E}\big[\exp\big(8|T - \mathbb{E}[T]|/Q_\ell\big)\big] \le C \qquad \text{and} \qquad \text{(b)}\ \ \mathrm{Var}[T] \le \omega_K Q_\ell^2\,,$$

with $\omega_K \to 0$ as $K \to \infty$, and, for $\ell = N$,

$$\text{(c)}\ \ \mathbb{E}[T] - \mu(N,y) \le Q_N/8\,.$$

They give (3.2) for K large.

4. Passage times between cubes

As in [AB26, Section 4], we extend the induction hypothesis (3.1) to bounds uniform over two cubes in the faces of a slab, and to the endpoints (ℓ, y) with $\ell \in [\kappa N, 3N]$, for $\kappa \in (0,1]$. A *cube* of side $\sigma > 0$ is the set of lattice points of $\{p\} \times (z + [0,\sigma)^{d-1})$, for $p \in \mathbb{Z}$ and $z \in \mathbb{R}^{d-1}$, when this set is nonempty; its center is the center of $\{p\} \times (z + [0,\sigma]^{d-1})$.

For an integer $s \geq 1$ and $t \in \mathbb{R}^{d-1}$, a pair of cubes $I \subseteq \{x \cdot e_1 = p\}$ and $I' \subseteq \{x \cdot e_1 = p+s\}$ has *slope* t if the difference of their centers lies within distance $2d^{1/2}$ of (s, st). For $\varrho > 0$, the *thickening* of I by ϱ along $(1,t)$ is the set of lattice points at distance at most $d^{1/2}$ from $\{u + q(1,t) : u \in I,\ |q| \leq \varrho\}$. These sets replace the cubes and boxes of [AB26, Section 4].

Lemma 4.1. *Let $b \geq 1$ and $s \in [N^{1/2}/16, N/2]$, let $t \in \mathbb{R}^{d-1}$ satisfy $|t - t_0| \leq \frac{4}{5}\, s^{\delta/100 - 1/4}$, let I be a cube of side at most $b(sQ_s)^{1/2}$ with center $\bar{u}$, and let $a \coloneqq \bar{u} - s(1,t)$ and $a' \coloneqq \bar{u} + s(1,t)$. If N is sufficiently large, depending only on K and b, then, for every subgradient ζ_t of μ at $(1,t)$,*

$$\sup_{u \in I} \big(T(a,u) - \zeta_t \cdot (u-a) \big) \leq \mathcal{O}_1(C(b) Q_s) \qquad \text{and} \qquad \sup_{u \in I} \big(T(u,a') - \zeta_t \cdot (a'-u) \big) \leq \mathcal{O}_1(C(b) Q_s)\,, \tag{4.1}$$

and the same holds with I replaced by its thickening by $s/64$ along $(1,t)$.

Proof. This is the chaining of [AB26, Lemma 4.1], with ζ_t in place of e_1. Its points $p(D)$ become $\bar{u}_j(u) - 2^{-j} s(1,t)$, where $\bar{u}_j(u)$ is the center of the dyadic subcube of I of generation j that contains u. While $2^{-j} s \geq C(b)$, consecutive points of this sequence differ by a vector whose slope lies within $(2^{-j-1}s)^{\delta/100 - 1/4}$ of t_0, by (3.4) and $\frac{4}{5} < 2^{1/4 - \delta/100}$. Hence (3.1) applies to them, and $Q_{2^{-j}s} \leq 2^{-2j/5} Q_s$, as in [AB26], since $\frac{1}{2} - \frac{\delta}{K} \geq \frac{2}{5}$.

In place of [AB26, (4.3)], (2.2) and (3.4) give, for consecutive points x, x' of generation j,

$$\mu(x' - x) - \zeta_t \cdot (x' - x) \leq C(b) 2^{-j} Q_s + C\,.$$

The last point is joined to u by a path of at most $C(b)$ edges. By (1.1), these paths have passage times at most $\mathcal{O}_1(C(b) \log(2+s))$, uniformly in u.

For a point of the thickening within distance $d^{1/2}$ of $u + q(1,t)$, with $u \in I$ and $|q| \leq s/64$, we add $q_j(1,t)$ to the point of generation j, where q_j is the center of the dyadic subinterval of $[-s/64, s/64]$ of generation j that contains q. For the second bound in (4.1), the points are $\bar{u}_j(u) + 2^{-j} s(1,t)$. □

Proposition 4.2 (Passage times between cubes)**.** *Let $b \geq 1$ and $\kappa \in (0,1]$, let s be an integer, and let I, I' be a pair of cubes of sides at most $b(sQ_s)^{1/2}$ with slope t, such that either $N^{1/2} \leq s \leq N/2$ and $|t - t_0| \leq \frac{4}{5}\, s^{\delta/100 - 1/4}$, or $\kappa N \leq s \leq 2N$ and $|t - t_0| \leq \frac{3}{2}\, s^{\delta/100 - 1/4}$. Let P be the slab between the hyperplanes of I and I'. If N is sufficiently large, depending only on K, b and κ, then*

$$\sup_{u \in I,\, v \in I'} |T_P(u,v) - \mu(v-u)| \leq \mathcal{O}_1(C(b) Q_s)\,, \tag{4.2}$$

the same holds for T, and, for T, it holds also with I and I' replaced by their thickenings by $s/1024$ along $(1,t)$.

Proof for $s \leq N/2$. The proof of [AB26, Proposition 4.2] applies to T, with ζ_t in place of e_1. Let $\bar{u}$ and $\bar{v}$ be the centers of I and I'. The points of $\mathbb{R}e_1$ in that proof become $\bar{u} \pm \frac{s}{16}(1,t)$ and $\bar{v} \pm \frac{s}{16}(1,t)$. The vectors between these points have first coordinates $9s/8$ and $7s/8$ and slopes within $\frac{4}{5}\, s^{\delta/100 - 1/4} + C/s < (9s/8)^{\delta/100 - 1/4}$ of t_0, so (3.1) applies to them.

Since $(sQ_s)^{1/2} \leq 8(\frac{s}{16} Q_{s/16})^{1/2}$, Lemma 4.1 with $s/16$ and $8b$ in place of s and b bounds the four other terms. By (2.2) and (3.4), the terms $\zeta_t \cdot (x' - x)$ and $\mu(x' - x)$ of these bounds add up to $\zeta_t \cdot (v - u)$ up to CQ_s, and $0 \leq \mu(v-u) - \zeta_t \cdot (v-u) \leq C(b) Q_s$. The thickenings by $s/1024$ are those of Lemma 4.1 with $s/16$ in place of s.

For T_P, we replace [AB26, (2.2)] by the following bound. By stationarity, we may assume that $|\bar{u}| \leq d^{1/2}$, so that I and I' lie in $[-N^2, N^2]^d$. Let $S \geq \log N$ be the least integer such

that $\mathcal{G}_N(S)$ occurs, so that $S \le \log N + 1 + \mathcal{O}_2(C)$. If $s \ge CS^2$, Lemma 2.5(1) applies, since the slopes of the vectors $v-u$ lie within $2s^{\delta/100-1/4} \le 3N^{-1/9}$ of t_0. Otherwise, Lemma 2.3(2) gives $T_P \le CS^2$. Hence

$$0 \le \sup_{u\in I,\, v\in I'} \big(T_P(u,v) - T(u,v)\big) \le CS^2 \le C\log^2 N + \mathcal{O}_1(C)\,. \tag{4.3}$$ □

For integers $M \ge 1$ and $s \ge N^{1/2}$, and $t \in \mathbb{R}^{d-1}$ with $Mst \in \mathbb{Z}^{d-1}$, we cut the faces of the slabs of (2.8) into cubes of side $w := \lceil (sQ_s)^{1/2} \rceil$: for $0 \le i \le M$ and $j \in \mathbb{Z}^{d-1}$, let $I_i(j)$ be the set of lattice points of $\{is\} \times (\lfloor ist \rfloor + w(j + [0,1)^{d-1}))$, with the floor taken coordinatewise.

Let F be the minimum (2.8) for M slabs of width s, from 0 to (Ms, Mst), let $x_0, \dots, x_M$ be the lexicographically least chain attaining it, and let J_i be the index with $x_i \in I_i(J_i)$, so that $J_0 = J_M = 0$.

For N large, if $u \in I_{i-1}(j)$ and $v \in I_i(k)$ are consecutive points of a chain, then $8w|k-j| \le s^{3/4+\delta/100}$, by (3.4). For an array $Z_i(j,k)$, we write $Z_i(x_{i-1}, x_i) := Z_i(J_{i-1}, J_i)$.

Lemma 4.3. *If N is sufficiently large, depending only on K, then, for M, s and t as above with $M \ge 128$, $Ms \le 3N$ and $|t - t_0| \le 2(Ms)^{\delta/100-1/4}$,*

$$|F - \mu(Ms, Mst)| + cQ_s \sum_{i=1}^{M} |J_i - J_{i-1}|^2 \le CMQ_s + \mathcal{O}_1(CQ_s)\,. \tag{4.4}$$

Proof. The proof of [AB26, Lemma 4.3] applies, with $T^{(i)}(u,v) - \zeta_t \cdot (v-u)$ in place of $T^{(i)}(u,v) - s$, for a subgradient ζ_t of μ at $(1,t)$. For $8w|k-j| \le s^{3/4+\delta/100}$, the cubes $I_{i-1}(j)$ and $I_i(k)$ form a pair of sides at most $2(sQ_s)^{1/2}$, with a slope within $4/5\, s^{\delta/100-1/4}$ of t_0, since $2 \cdot 128^{\delta/100-1/4} + 1/8 \le 4/5$. Hence Proposition 4.2 for $T^{(i)}$ and (2.2), with $w^2 \ge sQ_s$, give

$$\Big(cQ_s|k-j|^2 - \inf_{u\in I_{i-1}(j),\, v\in I_i(k)} \big(T^{(i)}(u,v) - \zeta_t \cdot (v-u)\big)\Big)_+ \le \mathcal{O}_1(CQ_s)\,.$$

The upper bound on F uses the chain of the rounded points (is, ist). □

As in [AB26, (4.7)], (4.4) and [AB26, Lemma 2.4] give, for M, s, t and N as in Lemma 4.3 and independent random arrays $Z_1, \dots, Z_M$ of nonnegative random variables with $\mathbb{E}[(Z_i(j,k) - |j-k|^2)_+^{4d}] \le 1$ for all i, j, k,

$$\mathbb{E}\Big[\sum_{i=1}^{M} Z_i(x_{i-1}, x_i)\Big] \le CM\,. \tag{4.5}$$

The next lemma gives the bound (a) of Section 3.

Lemma 4.4. *Let $\kappa \in (0,1]$. If N is sufficiently large, depending only on K and κ, then, for every integer $\ell \in [\kappa N, 3N]$ and every $y \in \mathbb{Z}^{d-1}$ with $|y - \ell t_0| \le 7/4\, \ell^{3/4+\delta/100}$,*

$$|T(0,(\ell,y)) - \mu(\ell,y)| \le CQ_\ell + \mathcal{O}_1(Q_\ell/100)\,. \tag{4.6}$$

Proof. We follow [AB26, proof of Lemma 4.4], with $s := \lfloor \ell/M \rfloor$ for an integer $M \ge 128$, so that $Q_s \le M^{-2/5} Q_\ell$, as in [AB26]. We take M so large that $C_1 M^{-2/5} \le 1/200$, where C_1 is the constant of the term $\mathcal{O}_1(CQ_s)$ in (4.4).

Let $t := y/(Ms)$. Since $0 \le \ell - Ms < M$ and $|y - \ell t_0| \le 7/4 \ell^{3/4+\delta/100}$, we have $|t - t_0| \le 2(Ms)^{\delta/100-1/4}$ for N large, and it suffices to bound $T(0,x) - \mu(x)$ for $x := (Ms, y)$, up to $\mathcal{O}_1(CM)$.

The comparison of F with $T(0,x)$ by Lemma 2.5(2) holds only on $\mathcal{G} := \mathcal{G}_N(N^{\delta/1000})$, where it gives $F - T(0,x) \le CMN^{\delta/500}$. Off $\mathcal{G}$, we use $(\mu(x) - T(0,x))_+ \le \mathcal{O}_2(C\ell^{1/2})$, by [ADH17, Theorem 3.13] and $\mu(x) \le \mathbb{E}[T(0,x)]$, and $\mathbf{1}_{\mathcal{G}^c} \le \mathcal{O}_2(CN^{-\delta/1000})$. Their product is at most $\mathcal{O}_1(C\ell^{1/2} N^{-\delta/1000})$, which is much smaller than Q_ℓ since $K \ge 10^4$. □

Proof of Proposition 4.2 for $\kappa N \le s \le 2N$. The proof for $s \le N/2$ applies, with (4.6) in place of (3.1) for the vectors with first coordinates $9s/8$ and $7s/8$, which lie in $[\kappa N/2, 3N]$. Lemma 4.1 still applies with $s/16$ in place of s, since $3/2 \cdot 16^{\delta/100-1/4} < 4/5$. □

5. Transverse fluctuations and continuity in the endpoints

As in [AB26, Section 5], we bound the distance from the geodesics between two cubes to the segments between their endpoints, and the modulus of continuity of the passage times between the cubes. Throughout, $b \ge 1$, $\kappa \in (0,1]$, $\ell \in [\kappa N, N]$ is an integer, and I, I' is a pair of cubes of sides at most $b(\ell Q_\ell)^{1/2}$, as in Section 4 with $s = \ell$, with a slope t such that $|t - t_0| \le \ell^{\delta/100-1/4}$. We fix a subgradient ζ_t of μ at $(1,t)$. Let $\bar u$ and $\bar v$ be the centers of I and I', and let P be the slab between their hyperplanes.

Lemma 5.1 (Transverse fluctuations)**.** *Let $p \ge 1$, and let*

$$R := 1 + (\ell Q_\ell)^{-1/2} \sup_{u \in I,\, v \in I'} \sup_{\pi} \sup_{x \in \pi} \operatorname{dist}(x, [u,v])\,,$$

where x ranges over the vertices of π, and π ranges either over the geodesics from u to v or over the geodesics from u to v in P. If N is sufficiently large, depending only on K, b, κ and p, then $\mathbb{E}[R^p] \le C(b,p)$ for both choices of π.

Lemma 5.2 (Continuity in the endpoints)**.** *For every $\eta > 0$, there exists $h > 0$, depending only on b and η, such that, if N is sufficiently large, depending only on K, b, κ and η, then*

$$\mathbb{E}\Big[\sup_{\substack{u,\widetilde u \in I,\ v,\widetilde v \in I' \\ |u-\widetilde u|,\, |v-\widetilde v| \le h(\ell Q_\ell)^{1/2}}} \big| T(u,v) - T(\widetilde u, \widetilde v) - \zeta_t \cdot (v - u - \widetilde v + \widetilde u) \big|^{8d} \Big]^{1/8d} \le \eta Q_\ell\,. \tag{5.1}$$

Proof of Lemma 5.1. We follow [AB26, proof of Lemma 5.1]. In place of the two points of that proof, at distance of order n outside its box, we take the points $a := \bar u - \ell/16(1,t)$ and $a' := \bar v + \ell/16(1,t)$, which the confinement below allows, and $\zeta_t \cdot (a' - a)$ in place of the distance between them. By stationarity, we may assume that $|\bar u| \le d^{1/2}$. Let $T(\pi) := \sum_{e \in \pi} \tau_e$. The upper bound

$$T(a,u) + T(\pi) + T(v,a') \le \zeta_t \cdot (a' - a) + \mathcal{O}_1(C(b) Q_\ell)\,,$$

uniform in u, v and π, follows from Lemma 4.1 with $\ell/16$ and $8b$ in place of s and b, Proposition 4.2 and (2.2). The proof changes in three places.

First, [AB26] confines the curves to a box by the bound on the density. Here, we work on the event $\mathcal{G}_N(\log N)$, which is contained in $\mathcal{G}_N(C \log N)$. On it, by Lemma 2.5(1) with $S = \log N$, the geodesics in P have passage time at most $T(u,v) + C \log^2 N$, so Lemma 2.4 applies to them, and to the geodesics, with $S = C \log N$. As in the proof of Lemma 2.5(1), it confines the vertices of the paths π to distance $C \log^2 N$ from P along e_1, and to distance at most

$$C\big(b(\ell Q_\ell)^{1/2} + \ell^{3/4} (\log N)^{1/2} + \ell^{1 + (\delta/100 - 1/4)(1+\delta/2)} \log N \big)$$

from the line $\bar u + \mathbb{R}(1,t)$. For N large, this is at most $\ell^{3/4+\delta/200}$, since $K \ge 10^4$ and

$$1 - \big(\tfrac14 - \tfrac{\delta}{100}\big)\big(1 + \tfrac{\delta}{2}\big) < \tfrac34\,.$$

Second, the boxes $B_{j,k}$ of [AB26] are replaced as follows. Fix an integer $\beta \ge 1$ with $\beta(\ell Q_\ell)^{1/2} \le \ell^{3/4+\delta/200}$. The vertices at distance between $\beta(\ell Q_\ell)^{1/2}$ and $(\beta+1)(\ell Q_\ell)^{1/2}$ from the line $\bar u + \mathbb{R}(1,t)$, and at distances in $[\ell/16 - C\log^2 N, 17\ell/16 + C \log^2 N]$ from a and a' along e_1, are covered by at most $C(b)\beta^d$ thickenings by $c_0 \ell/\beta$, for a small constant $c_0 > 0$ fixed below, along $(1,t)$ of cubes of side $(\ell Q_\ell)^{1/2}$, at distances in $[\ell/32, 9\ell/8]$ from a and a' along e_1.

By (2.2) and (3.4), the vertices x of these thickenings satisfy

$$\mu(x - a) + \mu(a' - x) \ge \zeta_t \cdot (a' - a) + c\beta^2 Q_\ell - C Q_\ell\,.$$

For each of these cubes, let I'' be the cube of side $2(\ell Q_\ell)^{1/2}$ with the same center.

Let t_1 and t_2 be the slopes of the vectors from a to this center and from this center to a'. Then $|t_1 - t|, |t_2 - t| \le C\beta(Q_\ell/\ell)^{1/2}$, which is much smaller than $\ell^{\delta/100-1/4}$, so $|t_1 - t_0|, |t_2 - t_0| \le 3/2(9\ell/8)^{\delta/100-1/4}$. Hence Proposition 4.2, with $\kappa/32$ in place of κ, applies to the pairs $\{a\}, I''$ and $I'', \{a'\}$, whose slopes are t_1 and t_2.

For c_0 small, the thickening lies in the thickenings of I'' along $(1, t_1)$ and along $(1, t_2)$ that this proposition covers. We sum the tails of its $\mathcal{O}_1$ terms over the thickenings and over β, as in the proof of [AB26, (5.3)]. This gives $R\mathbf{1}_{\mathcal{G}_N(\log N)} \le \mathcal{O}_2(C(b))$.

Third, off $\mathcal{G}_N(\log N)$, the bound on the number of edges of geodesics in Section 2 gives $R \le C\ell + \mathcal{O}_1(C\log\ell)$, and $\mathbf{1}_{\mathcal{G}_N(\log N)^c} \le \mathcal{O}_2(C/\log N)$, so Hölder's inequality bounds $\mathbb{E}[R^p \mathbf{1}_{\mathcal{G}_N(\log N)^c}]$ by 1 for N large. □

Proof of Lemma 5.2. We follow [AB26, proof of Lemma 5.2], with $\gamma \in (0, 1/128)$ and $h := \gamma^3$, as there; the bound $\gamma < 1/128$ makes Proposition 4.2 apply with $s = \lceil \gamma\ell \rceil$. There are two substitutions: the event $\{R > 1/\gamma\}$, for R as in Lemma 5.1 with the geodesics from u to v, in place of E; and $\phi(z) := \mu(z) - \zeta_t \cdot z$ in place of $|z| - e_1 \cdot z$.

Instead of the subslabs S_1 and S_2, we reroute a geodesic from u to v at two of its vertices: the first vertex x with $x \cdot e_1 \ge u \cdot e_1 + \gamma\ell$, and the last vertex x' with $x' \cdot e_1 \le v \cdot e_1 - \gamma\ell$. Since the first coordinate changes by at most one along each edge, $x \cdot e_1 = u \cdot e_1 + \lceil \gamma\ell \rceil$. Since $2\lceil \gamma\ell \rceil < \ell$, the vertex x precedes x'. Hence

$$T(\widetilde{u}, \widetilde{v}) - T(u, v) \le T(\widetilde{u}, x) - T(u, x) + T(x', \widetilde{v}) - T(x', v)\,,$$

in place of the display of [AB26] with T_{S_1} and T_{S_2}, and D is defined with $P = \mathbb{R}^d$ only, and with the hyperplanes $\{x \cdot e_1 = u \cdot e_1 + \lceil \gamma\ell \rceil\}$ and $\{x \cdot e_1 = v \cdot e_1 - \lceil \gamma\ell \rceil\}$ in place of H_1 and H_2.

The pairs (u, x), $(\widetilde{u}, x)$, (x', v) and $(x', \widetilde{v})$ lie in at most $C(b)\gamma^{-C}$ pairs of cubes as in Proposition 4.2, with $s = \lceil \gamma\ell \rceil$ and sides $(\gamma\ell Q_{\gamma\ell})^{1/2}$. The bound $Q_{\lceil \gamma\ell \rceil} \le 2\gamma^{2/5} Q_\ell$, which holds since $1/2 - \delta/K \ge 2/5$, replaces $Q_{\gamma n} \le \gamma^{2/5} Q_n$.

By (2.2) and (3.4), $\phi \le C(b)\gamma^{-3} Q_\ell$ at the points within distance $\gamma^{-1}(\ell Q_\ell)^{1/2}$ of $x - u$ and of $v - x'$. Since ϕ is convex and nonnegative, moving the vectors $x - u$ and $v - x'$ by at most $h(\ell Q_\ell)^{1/2}$ changes ϕ by at most $C(b)\gamma Q_\ell$. For the supremum Z in (5.1), this gives

$$\mathbb{E}\big[Z^{8d} \mathbf{1}_{\{R \le 1/\gamma\}}\big]^{1/8d} \le C(b)\big(\gamma + \gamma^{2/5}\log(2/\gamma)\big) Q_\ell\,.$$

Proposition 4.2, (2.2) and Lemma 5.1 with $p = 16d$ give $\mathbb{E}[Z^{8d} \mathbf{1}_{\{R > 1/\gamma\}}]^{1/8d} \le C(b)\gamma Q_\ell$. □

6. A logarithmic gain of variance

As in [AB26, Section 6], we prove the bound (b) of Section 3. We let ω_K and ω_L denote numbers depending only on K, respectively on L, that tend to zero as $K \to \infty$, respectively $L \to \infty$, and may change from line to line.

Let $\ell' \in [N/8, N]$ be an integer and $y' \in \mathcal{D}(\ell')$. We first replace ℓ' by a multiple of K. Let $\ell := K\lfloor \ell'/K \rfloor$. By (2.4), some $y \in \mathcal{D}(\ell)$ satisfies $|y - \ell y'/\ell'| \le 2(d-1)^{1/2}$, and then $|T(0, (\ell', y')) - T(0, (\ell, y))| \le \mathcal{O}_1(CK)$. From now on, $\ell = Kn \in [N/9, N]$ for an integer n, the vector $y \in \mathcal{D}(\ell)$ is fixed, and $t := y/\ell$. The objects w, $I_i(j)$, F, x_i and J_i are as in Section 4, with $M = K$ and $s = n$.

The comparison [AB26, (2.3)] of F with the passage time is replaced by

$$\mathbb{E}\big[(F - T(0, (\ell, y)))^2\big]^{1/2} \le CK\log^2 N\,, \tag{6.1}$$

which follows from Lemma 2.5(2) with $S = \log N$ on $\mathcal{G}_N(\log N)$, and off it from Hölder's inequality and the moments of the bound

$$F \le \sum_{i=1}^{K} T^{(i)}\big((i-1)n(1,t), in(1,t)\big)\,.$$

We shear the blocks along the slope t. For $j \in \mathbb{Z}^{d-1}$, the *cell* j of the ith slab is the set of its edges whose midpoints (q,h), with $q \in \mathbb{R}$ and $h \in \mathbb{R}^{d-1}$, satisfy $h - qt \in w(j + [0,1)^{d-1})$. For an integer $L \ge 1$ and vectors $a_i \in \{0,\dots,L-1\}^{d-1}$, the *blocks* of the ith slab are the unions of the cells $j + \{0,\dots,L-1\}^{d-1}$ for $j \in a_i + L\mathbb{Z}^{d-1}$.

In place of the tubes of [AB26, Section 6], we use the following variables. For $1 \le i \le K$ and $j,k \in \mathbb{Z}^{d-1}$ with $8w|k-j| \le n^{3/4+\delta/100}$, let $R_i(j,k) := CR$, where R is the random variable of Lemma 5.1 for the cubes $I_{i-1}(j)$ and $I_i(k)$, whose slope lies within $n^{\delta/100-1/4}$ of t_0 as in the proof of Lemma 4.3, and the geodesics in the ith slab, with $b = 2$ and $\kappa = 1/(9K)$, and let $R_i(j,k) := 0$ for the other j,k. We take C so large that every edge of these geodesics lies in a cell with an index z such that $\operatorname{dist}(z,[j,k]) \le R_i(j,k)$, where $[j,k] \subseteq \mathbb{R}^{d-1}$ is the segment between the indices.

We use the resampling τ^G of a block G given a random variable σ_G determined by $\tau|_G$, and the quantities $T^{(i),G}$, F^G and $Y_i(j,k)$, of [AB26, Section 6], with the weights τ in place of the noise ω, and with $Y_i(j,k) := 0$ unless $8w|k-j| \le n^{3/4+\delta/100}$. Since $T^{(i),G}(u,v) > T^{(i)}(u,v)$ only if every geodesic from u to v in the ith slab has an edge in G, the term of G in $Y_i(j,k)$ vanishes unless a cell of G has index z with $\operatorname{dist}(z,[j,k]) \le R_i(j,k)$.

Lemma 6.1. *If N is sufficiently large, depending only on K, then*

$$\mathbb{E}\Big[\sum_{i=1}^{K} R_i(x_{i-1},x_i)^2\Big] \le CK\,, \tag{6.2}$$

and, for every integer $L \ge 1$, every choice of the vectors a_i and of the σ_G, and all i,j,k,

$$\mathbb{E}\big[Y_i(j,k)^{16d}\big]^{1/16d} \le C(1+|j-k|) \qquad \text{and} \qquad \sum_{G} \mathbb{E}\big[(F^G - F)^2\big] \le CKQ_n^2\,, \tag{6.3}$$

where the sum is over the blocks of all K slabs.

Proof. The proof of [AB26, Lemma 6.1] applies, with the cells of this section, with (4.5) in place of [AB26, (4.7)], and with the moment bound $\mathbb{E}[R_i(j,k)^{128d^2}] \le C$ of Lemma 5.1 in place of the Gaussian tail of $R_i(j,k)$.

By Hölder's and Markov's inequalities, the L^{16d} norm of the term of the cell z in the bound of $Y_i(j,k)$ is at most $C(1+\operatorname{dist}(z,[j,k]))^{-4d}$, and

$$\sum_{z\in\mathbb{Z}^{d-1}} \big(1+\operatorname{dist}(z,[j,k])\big)^{-4d} \le C(1+|j-k|)\,.$$

The proof also gives

$$\sum_{G} \mathbb{E}\big[(F^G - F)^2\big] \le 2Q_n^2\,\mathbb{E}\Big[\sum_{i=1}^{K} Y_i(x_{i-1},x_i)\Big]\,. \tag{6.4}$$

□

We fix a subgradient ζ_t of μ at $(1,t)$. For a block G of the ith slab, let j be such that G is the union of the cells $j + \{0,\dots,L-1\}^{d-1}$, and let Γ_G be the set of the pairs (u,v) of lattice points of the faces $\{x\cdot e_1 = (i-1)n\}$ and $\{x\cdot e_1 = in\}$ such that each of u and v has coordinates (q,h) with

$$h - qt \in w\big(z + [0,1)^{d-1}\big) \qquad \text{for some } z \in j + \{\lceil L^{1/2}\rceil,\dots,L-1-\lceil L^{1/2}\rceil\}^{d-1}.$$

Lemma 6.2. *Let $L \geq 16$ be an integer. There exists $\alpha_L > 0$, depending only on L, such that the following holds if N is sufficiently large, depending only on K and L. For every choice of the vectors a_i, every $i \in \{1, \dots, K\}$ and every block G of the ith slab, there exists a random variable X_G, which depends on the vectors $a_1, \dots, a_K$ only through G, is determined by $\tau|_G$, taking finitely many values, each a bounded function on Γ_G and each with probability at least α_L, such that*

$$\mathbb{E}\Big[\sup_{(u,v)\in\Gamma_G} \Big| \frac{T^{(i)}(u,v) - \zeta_t \cdot (v-u)}{Q_n} - X_G(u,v)\Big|^{8d}\Big]^{1/8d} \leq \omega_L \,. \tag{6.5}$$

Moreover, let $m \in \mathbb{Z}^{d-1}$ and $z := (0, wLm)$. For every block G of the ith slab and every $(u,v) \in \Gamma_G$, the value $X_{G+z}(u+z, v+z)$ computed from the weights $(\tau_{e-z})_e$ equals $X_G(u,v)$ computed from $(\tau_e)_e$.

Proof. Steps 1 and 2 of [AB26, proof of Lemma 6.2] change, since a block of edges has no passage time T_G of its own. Step 3 applies, with Ψ_G written as a function of $(u,v) \in \Gamma_G$, with the mesh parameter g in place of κ, and with the translations by the vectors of $\{0\} \times wL\mathbb{Z}^{d-1}$, under which the mesh is invariant, in place of the translations orthogonal to e_1 of [AB26].

In Step 1, we condition on the weights in G. Let $\Phi := (T^{(i)}(u,v) - \zeta_t \cdot (v-u))/Q_n$ on Γ_G, let $\Psi := \mathbb{E}[\Phi \mid \tau|_G]$, which is determined by $\tau|_G$, and let Φ' be Φ computed with τ replaced outside G by an independent copy, so that $\Psi = \mathbb{E}[\Phi' \mid \tau]$.

If $\Phi' \neq \Phi$ somewhere on Γ_G, then $R_i(j', k') \geq cL^{1/2}$, computed for τ or for the weights of Φ', for one of at most $CL^{2(d-1)}$ pairs of cubes $I_{i-1}(j')$ and $I_i(k')$ that cover Γ_G. By Lemma 5.1 with $p = 8d$, this has probability at most $CL^{2(d-1)}L^{-4d}$. In $\Phi - \Phi'$ the terms $\zeta_t \cdot (v-u)$ cancel, so Proposition 4.2, applied to these pairs of cubes, and (1.1) give $\sup_{\Gamma_G} |\Phi - \Phi'| \leq \mathcal{O}_1(C\log(2+L))$. With the Hölder and conditional Jensen inequalities, this gives $\mathbb{E}[\sup_{\Gamma_G} |\Phi - \Psi|^{8d}]^{1/8d} \leq \omega_L$.

In Step 2, we discretize Ψ instead of T_G. For $g \in (0,1]$, let (u_g, v_g) be a point of a mesh of Γ_G of spacing $g(nQ_n)^{1/2}$ with $|u - u_g|, |v - v_g| \leq g(nQ_n)^{1/2}$. We choose the mesh so that translating G, u and v by a vector of $\{0\} \times wL\mathbb{Z}^{d-1}$ translates (u_g, v_g) by the same vector. Let

$$\Psi_G(u,v) := g\big\lfloor g^{-1} \max\{-L^4, \min\{L^4, \Psi(u_g, v_g)\}\}\big\rfloor \,,$$

which is determined by $\tau|_G$ and takes at most $(2L^4/g + 2)^{(CL/g)^{2(d-1)}}$ values. By Proposition 4.2 and (2.2), $\sup_{\Gamma_G} |\Phi| \leq CL^2 + \mathcal{O}_1(C\log(2+L))$. By (4.3), (5.1) with $b = 2L$ and $\kappa = 1/(9K)$ holds for $T^{(i)}$ with $2\eta Q_n$ in place of ηQ_n. By the conditional Jensen inequality, Ψ inherits both bounds. Hence $\mathbb{E}[\sup_{\Gamma_G} |\Psi - \Psi_G|^{8d}]^{1/8d} \leq \omega_L$ for g small, depending on L. □

From now on, the X_G, for the blocks G of all K slabs, are as in Lemma 6.2, and F_m, F_m^G, $\overline{F}$, f and $\mathbb{E}_G[f]$ are as in [AB26, Section 6], with the endpoints 0 and (ℓ, y) of F_m shifted by $mLwe_2$.

Lemma 6.3. *Let $L \geq 16$ be an integer, and let τ^G be the resampling of G with $\sigma_G = X_G$. If N is sufficiently large, depending only on K and L, then*

$$\frac{1}{L^{K(d-1)}} \sum_a \sum_G \mathbb{E}\big[(F^G - F)^2\big] \leq \omega_L K Q_n^2 \qquad \text{and} \qquad \frac{1}{L^{K(d-1)}} \sum_a \mathbb{E}\big[\mathrm{Var}[\overline{F} \mid (X_G)_G]\big] \leq \omega_L K Q_n^2 \,, \tag{6.6}$$

where $a = (a_1, \dots, a_K)$ ranges over $(\{0, \dots, L-1\}^{d-1})^K$; the blocks, the X_G and τ^G depend on a.

Proof. The proof of [AB26, Lemma 6.3] applies, with (4.5), (6.3) and (6.4) in place of [AB26, (4.7), (6.2) and (6.3)] and $\zeta_t \cdot (v-u)$ in place of n.

Let G be the block that contains the cell j, and let j_G be such that G is the union of the cells $j_G + \{0, \dots, L-1\}^{d-1}$. The event E of that proof becomes the event that $R_i(j,k) > L^{1/2}$, or that some $z \in \mathbb{Z}^{d-1}$ with $|z - j| \leq C(1 + |j-k| + L^{1/2})$ lies outside $j_G + \{\lceil L^{1/2} \rceil, \dots, L - 1 - \lceil L^{1/2} \rceil\}^{d-1}$. Lemma 5.1 and Markov's inequality give $\mathbb{P}[E] \leq C(1 + |j-k| + L^{1/2})/L$. □

Lemma 6.4. *Let $L \ge 16$ be an integer. If N is sufficiently large, depending only on K and L, then, for every choice of the vectors a_i,*

$$\sum_G \mathbb{E}\big[(f - \mathbb{E}_G[f])^2\big] \le CKQ_n^2 \qquad \text{and} \qquad \sum_G \mathbb{E}\big[|f - \mathbb{E}_G[f]|\big]^2 \le CK^{-1/6}KQ_n^2 \,.$$

Proof. The proof of [AB26, Lemma 6.4] applies, with (6.3) in place of [AB26, (6.2)], and with the cube indices of the minimizing chains in place of their points. The cells with indices z such that $\operatorname{dist}(z, [J_{i-1}, J_i]) \le R_i(J_{i-1}, J_i)$ lie in at most

$$C\big(1 + (|J_i - J_{i-1}| + R_i(J_{i-1}, J_i))/L\big)$$

of the blocks $G + m'Lwe_2$ with $m' \in \mathbb{Z}$. By (4.4) and (6.2), the squares of these numbers, summed over i, are at most CK in mean. □

Proposition 6.5 (The gain)**.** *If K is sufficiently large and N is sufficiently large depending on K, then*

$$\operatorname{Var}[T(0, (\ell', y'))] \le \omega_K Q_{\ell'}^2 \qquad \text{for every integer } \ell' \in [N/8, N] \text{ and every } y' \in \mathcal{D}(\ell').$$

Proof. The proof of [AB26, Proposition 6.5] applies, with Lemmas 6.2–6.4 in place of [AB26, Lemmas 6.2–6.4], and gives $\operatorname{Var}[\overline{F}] \le \omega_K K Q_n^2$. In its comparison of $\overline{F}$ with the passage time, the translation vectors $mLwe_2$ have length at most $2K^{-1/24+\delta/(2K)}(\ell Q_\ell)^{1/2}$, since $L \le K^{1/24}$ there. Hence Lemma 5.2, for cubes in the hyperplanes $\{x \cdot e_1 = 0\}$ and $\{x \cdot e_1 = \ell\}$, and (6.1) give

$$\mathbb{E}\big[(\overline{F} - T(0, (\ell, y)))^2\big]^{1/2} \le \omega_K Q_\ell \,.$$

Finally, $KQ_n^2 = K^{2\delta/K} Q_\ell^2 \le 2Q_\ell^2$. □

7. The mean and the induction step

The bound (c) of Section 3, at $\ell = N$, follows from the next lemma, which replaces [AB26, Lemma 7.1].

Lemma 7.1 (Nonrandom fluctuations)**.** *If N is sufficiently large, depending only on K, then, for every integer $s \in [N/8, N/4]$ such that $\sup_{y \in \mathcal{D}(s)} \operatorname{Var}[T(0, (s, y))/Q_s]$ is at most c, and every $y_0 \in \mathbb{Z}^{d-1}$ with $|y_0 - st_0| \le sN^{\delta/100 - 1/4}$,*

$$\mathbb{E}[T(0, (s, y_0))] - \mu(s, y_0) \le Q_s/100 \,.$$

Proof. We follow [AB26, proof of Lemma 7.1], for s slabs of width s from 0 to (s^2, sy_0), in the notation of Section 4 with $M = s$ and $t := y_0/s$. Let ζ_t be a subgradient of μ at $(1, t)$, so that $\zeta_t \cdot (s, y_0) = \mu(s, y_0)$, and suppose that $\mathbb{E}[T(0, (s, y_0))] - \mu(s, y_0) > Q_s/100$.

By Jensen's inequality and Lemma 5.2 with $b = 2$, $\kappa = 1/8$ and $\eta = 1/200$, for $z \in \mathbb{Z}^{d-1}$ with $|z| \le h(sQ_s)^{1/2}$,

$$\mathbb{E}[T(0, (s, y_0 + z))] - \zeta_t \cdot (s, y_0 + z) \ge Q_s/200 \,.$$

For $h(sQ_s)^{1/2} < |z| \le s^{3/4+\delta/100}/4$, the bound $\mathbb{E}[T] \ge \mu$, (2.2) and (3.4) bound the same difference below by $c|z|^2/s - ch^2 Q_s/4$, where c is the constant of (2.2). Hence, for a constant $c_1 > 0$,

$$\mathbb{E}[T(0, (s, y_0 + z))] - \zeta_t \cdot (s, y_0 + z) \ge c_1 Q_s + c_1 |z|^2/s \qquad \text{for } |z| \le s^{3/4+\delta/100}/4. \tag{7.1}$$

For $u \in I_{i-1}(j)$ and $v \in I_i(k)$ with $8w|k - j| \le s^{3/4+\delta/100}$, we have $v - u = (s, y_0 + z)$ with $|z - w(k - j)| \le wd^{1/2}$, so (7.1) and $w^2 \ge sQ_s$ give

$$\mathbb{E}[T(u, v)] - \zeta_t \cdot (v - u) \ge c_1 Q_s + c_1 Q_s (|k - j| - d^{1/2})_+^2 \,.$$

This replaces the lower bound $(\mathbb{E}[T_s] - s)/(1 + \sup \rho)$ of [AB26].

Let $\theta \in (0,1]$. For j,k with $8w|k-j| \le s^{3/4+\delta/100}$, let

$$Z_i(j,k) := \sup_{u\in I_{i-1}(j),\, v\in I_i(k)} \big|T^{(i)}(u,v) - \mathbb{E}[T(u,v)]\big| / (\theta Q_s)\,,$$

and let $Z_i(j,k) := 0$ for the other j,k. If c in the hypothesis on the variance is small and N is large, depending on θ, then $\mathbb{E}[Z_i(j,k)^{4d}] \le 1$. This is the first claim of [AB26, proof of Lemma 7.1], $\mathbb{E}[X_i(j,k)^{4d}]^{1/4d} \le \theta$ for $X_i := \theta Z_i$, with θ in place of the κ there and $\mathbb{E}[T(u,v)]$ in place of $\mathbb{E}[T^{(i)}(u,v)]$. Its proof applies, with (4.6), (4.3) and (5.1) in place of [AB26, (4.4), (2.2) and (5.2)], and with two changes.

First, for $8w|k-j| \le s^{3/4+\delta/100}$, the pairs of cubes $I_{i-1}(j)$ and $I_i(k)$ have slopes within $s^{\delta/100-1/4}/4$ of t. Since $|t-t_0| \le (4s)^{\delta/100-1/4}$ and $4^{\delta/100-1/4} + 1/4 < 1$, the differences $v-u$ of their points are (s,y) with $y \in \mathcal{D}(s)$, so the hypothesis on the variance applies to $T(u,v)$.

Second, we apply (5.1), with $b=2$ and $\kappa = 1/8$, to each of these pairs, with its own slope t' and a subgradient $\zeta_{t'}$ of μ at $(1,t')$. Subtracting $\zeta_{t'}\cdot(v-u)$ from $T(u,v)$ and from $\mathbb{E}[T(u,v)]$ does not change their difference. This replaces the subtraction of $|v-u|$ in [AB26].

Since $Ms = s^2 > 3N$, Lemma 4.3 and (4.5) do not apply here; in place of [AB26, (4.7)], we use the quadratic term in (7.1) and [AB26, Lemma 2.4]. Since $c_1(x-d^{1/2})_+^2 \ge \theta(x^2-2d)$ for $x \ge 0$ if $\theta \le c_1/2$, summing along the minimizing chain gives

$$F - s\mu(s,y_0) \ge (c_1 - 2d\theta)sQ_s - \theta Q_s \sup_{k_1,\dots,k_s} \Big\{ \sum_{i=1}^{s} Z_i(k_{i-1},k_i) - \sum_{i=1}^{s} |k_i - k_{i-1}|^2 \Big\}\,,$$

and [AB26, Lemma 2.4] gives $\mathbb{E}[F] - s\mu(s,y_0) \ge (c_1 - C\theta)sQ_s$.

The upper bound $\mathbb{E}[F] - s\mu(s,y_0) \le Cs\log^2 s$ follows as in [AB26], from Lemma 2.5(2), with s^2 in place of N, $n=s$ and $S = \log(s^2)$, and from (2.5) at (s^2, sy_0). Off $\mathcal{G}_{s^2}(\log(s^2))$, Hölder's inequality and the moments of the bound

$$F \le \sum_{i=1}^{s} T^{(i)}\big((i-1)(s,y_0), i(s,y_0)\big)$$

bound the contribution by 1. Since $\log^2 s$ is much smaller than Q_s, this contradicts the lower bound for θ small. □

Proof of Proposition 3.1. As in [AB26, proof of Proposition 3.1], we fix K large. By Proposition 6.5, for N large depending on K, the variance $\mathrm{Var}[T(0,(s',y'))/Q_{s'}]$ is as small as Lemma 7.1 requires, for all integers $s' \in [N/8, N]$ and $y' \in \mathcal{D}(s')$. By (4.6), the bound (a) holds with a constant independent of K, and the Cauchy–Schwarz inequality combines it with (b) into $|T - \mathbb{E}[T]| \le \mathcal{O}_1(Q_N/4)$, for $T := T(0,(N,y))$ and every $y \in \mathcal{D}(N)$.

Let $s := \lfloor N/4 \rfloor$. By (2.4), there is y_0 as in Lemma 7.1 with $|y_0 - sy/N| \le 2(d-1)^{1/2}$, and a path of at most C edges joins $(4s, 4y_0)$ to (N,y). Lemma 7.1 and subadditivity through the points $k(s,y_0)$ for $0 \le k \le 4$ give

$$0 \le \mathbb{E}[T] - \mu(N,y) \le 4\big(\mathbb{E}[T(0,(s,y_0))] - \mu(s,y_0)\big) + C \le Q_N/8\,.$$

Hence (3.2) follows by (1.2), and all the thresholds on N depend only on K. □

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